

Siva Nagendra Savarapu

Senior Pipeline Engineer · Generative AI & 3D Systems · OpenUSD · Unreal Engine · Python/C++

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SUMMARY

Senior Pipeline and Software Engineer with 10+ years building production infrastructure, AI systems, and real-time 3D workflows for leading VFX, game, and technology companies including Bezi, Epic Games, Sony Pictures Imageworks, Animal Logic, and MPC. Strong background in OpenUSD, Unreal Engine, Python/C++, TypeScript, Rust, agentic AI systems, and DCC integration. Currently building AI-agent infrastructure for Unreal Engine game development, including agentic tool discovery, virtual-file-system skills, evaluation systems, and cross-platform plugin delivery. Seeking senior roles across pipeline engineering, ML/AI, and technical AI for 3D, VFX, and game production.

EXPERIENCE



Senior Software Engineer - Bezi Inc

📅 May 2026 - Present · 📍 Vancouver, Canada

- Building an **Unreal Engine AI agentic harness for game development**, enabling agents to discover, reason about, and execute UE tools and actions through production-ready workflows.
- Authored a new **skills system backed by virtual file systems**, creating a scalable foundation for packaging, loading, and orchestrating agent capabilities.
- Implemented **agentic search and discovery systems** for Unreal Engine tools and actions, using LLM-as-judge workflows and industry-standard evaluation loops to measure relevance, correctness, and agent performance.
- Built CI systems to package and distribute **Unreal Engine plugins** across target platforms and multiple engine versions.
- Developed across **TypeScript, Rust, and C++ libraries**, integrating AI infrastructure with native Unreal Engine tooling and plugin ecosystems.



Senior Pipeline Engineer - Epic Games

📅 May 2025 - Mar 2026 · 📍 Vancouver, Canada

- Worked on a team building the **Fortnite** asset pipeline spanning Modelling, Texturing, Rigging, and Animation, engineering Python and C++ tools across Maya and Unreal Engine including a Rig Browser via OpenAssetIO, Maya to UE animation import/export, and ShotGrid production tracking automation to accelerate asset creation and eliminate manual handoff overhead across all departments.
- Designed and deployed **Model Context Protocol (MCP) servers for Maya and Unreal Engine**, integrating Claude Code into Epic's live production pipeline, among the earliest production MCP deployments at a major game studio.
- Engineered a **C++ Unreal Engine Runtime prototype** combining Claude (Anthropic Agent SDK) with UE MCP, Gemini/Meshy/Tripo for mesh generation, Depth Pro for depth estimation, Grounding DINO for object detection, and SAM2 for segmentation, delivering an end-to-end agentic 3D scene construction system from a single user prompt or image with mesh placement via a custom raycasting algorithm and GLTF plugin.
- Contributed to **GenStudio**, an internal generative AI product integrating Gemini, OpenAI image models, Meshy, Tripo, and Rodin mesh generation models for AI-assisted 3D asset creation workflows.
- Built a **RAG-powered pipeline assistant** that auto-generates visual workflow graphs (analogous to UE Blueprints) from natural language queries, with a custom RAG backend built on internal pipeline documentation.
- Implemented **C2PA content provenance workflows** across GenStudio, ensuring cryptographic content credential tracking for all AI-generated image outputs in compliance with emerging content authenticity standards.

 **Senior Software Engineer - Sony Pictures Imageworks** 📅 Jul 2023 - May 2025 · 📍 Vancouver, Canada

- Engineered an **OpenUSD and Alembic-based asset transfer pipeline from Houdini to Unreal Engine**, preserving textures, materials, and animation data through the full handoff, significantly accelerating the real-time layout process by eliminating manual re-assembly of assets inside UE.
- Built and maintained OpenUSD pipelines for VFX and Animation productions using Python, C++, and PySide6, covering asset and shot workflows across Modelling, Layout, and Animation departments.
- Developed DCC integrations across Maya, Houdini, and Unreal Engine ensuring pipeline data interoperability and compatibility across departments and studios.
- Designed modular show-specific tools in PySide6 and web frameworks deployed inside Maya and Unreal Engine based on production requirements.
- Contributed to productions including **Fantastic Four: First Steps** and **KPop Demon Hunters**.

 **Core Assets Pipeline Technical Director - Animal Logic** 📅 Aug 2021 - May 2023 · 📍 Sydney, Australia

- Developed and maintained core Asset Pipeline systems using Pixar's USD, Houdini, Maya, and Python, including USD schema extensions and pipeline plugin development.
- Designed UsdShade material workflows to improve USD data read/write fidelity; improved USD Assembly pipeline for environment workflows and asset validation systems.
- Contributed heavily to USD texture/material workflows for the **ALab public dataset release**.
- Enhanced Houdini UX for artists via Python and PySide6 tooling, reducing friction in day-to-day pipeline interactions.

 **Technical Director - MPC Film** 📅 Nov 2018 - Jul 2021 · 📍 Bangalore, India

- Built a Real-Time Asset presentation pipeline in Unreal Engine and developed MPC's core production pipeline tools using Python and PySide6, supporting artists across Rigging, Groom, Lighting, and FX departments.
- Built ION containers to manage reproducible software environments across shows.
- Contributed to productions including **WandaVision** and **Top Gun: Maverick**.

 **Texture & Lookdev Artist - Rhythm & Hues / Prana Studios** 📅 Sep 2015 - Nov 2018 · 📍 Mumbai, India

- Worked on **Hellboy (2019), 2.0 (2019)**, Fear the Walking Dead, and additional productions.
- Developed Lookdev and Lighting workflows with ImageEngine's Gaffer; trained the team in Gaffer, Substance Painter, and Mari.
- Worked on Unreal Engine and Unity for Previz and real-time VR projects.

EDUCATION

 **Bachelor's Degree, Computer Science** - Acharya Nagarjuna University, India · 2013 - 2016

SKILLS

 **Languages:** Python, C++, TypeScript, Rust, JavaScript, MEL

 **AI/ML:** PyTorch, TensorFlow, Computer Vision, Multimodal AI, LLMs, RAG, MCP, LLM-as-Judge, Agent Evals, Diffusion Models, Grounding DINO, Depth Pro, SAM2, Embeddings, C2PA

 **UI Frameworks:** PySide6/Qt6, web frameworks

 **Tools:** Git, GitHub, Docker, CI/CD, Virtual File Systems, GLTF

 **AI APIs & SDKs:** Anthropic Agent SDK, OpenAI API, Gemini API, Hugging Face, Meshy, Tripo, Rodin

 **Pipeline & DCCs:** OpenUSD, OpenAssetIO, ShotGrid, Maya, Houdini, Unreal Engine, Nuke, Gaffer, Alembic

 **Rendering & Real-Time:** UE5, PBR workflows, Arnold, Renderman, Substance Painter/Designer, Virtual Production